Microeconomics Midterm – Formula Sheet

ESADE BBA 26/27 · Topics 1–3

Percentages always use the **starting** (old) value. Δ means "change in" = new − old.

Topic 1 · Introduction to Economics

Opportunity cost

The value of the **next-best alternative** given up when you make a decision.

Example: Taxi job: lost salary 37,000 + lost interest 4% × 100,000 = 41,000.

Opportunity cost from a PPF

$$\text{OC of 1 unit of } X = \frac{\text{max. units of } Y}{\text{max. units of } X}$$

$$\text{OC of 1 unit of } Y = \frac{1}{\text{OC of } X}$$

The OC is the slope of the PPF (ignore the minus sign).

Example: France: 6 panels or 3 turbines ⇒ OC(turbine) $= \frac{6}{3} = 2$ panels.

PPF from time / resources

$$a \cdot X + b \cdot Y = \text{Total} \quad \Longrightarrow \quad Y = \frac{\text{Total}}{b} - \frac{a}{b} X$$

Example: HAL26: $3M + 12P = 120 \Rightarrow P = \frac{120}{12} - \frac{3}{12} M = 10 - 0.25\,M$

Comparative advantage & joint PPF

- Comparative advantage = the **lower** opportunity cost.
- Trade price lies **between** the two OCs.
- Joint PPF: start at max. of the vertical good. The producer with the **lowest OC of the horizontal good** switches first ⇒ **kink**. Then the other one.

Example: France/Germany: kink at (300 turbines, 1,800 panels), slopes −2 then −4.

HHI (market concentration)

$$\text{HHI} = s_1^2 + s_2^2 + s_3^2 + \ldots \qquad (s = \text{share in \%})$$

$$\text{Minimum with } n \text{ firms} = \frac{10{,}000}{n}$$

Range: close to 0 → 10,000 (monopoly).

Example: $80^2 + 15^2 + 5^2 = 6{,}650$; 3 firms: $\frac{10{,}000}{3} = 3{,}333$

Topic 2 · Market Analysis

Equilibrium

Set demand = supply: $Q_d = Q_s$

1. Solve for p^*.
2. Put p^* into **both** equations ⇒ Q^* (check!).

Example: $2000 - 2p = 200 + 3p \Rightarrow p^* = \frac{1800}{5} = 360$, $Q^* = 1280$

Percentage change

$$\%\Delta = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

$$\text{new value} = \text{old} \times \left(1 + \frac{\%\Delta}{100}\right)$$

Example: $\frac{425 - 250}{250} \times 100 = +70\%$; $200 \times (1 - 0.105) = 179$

Price elasticity of demand (PED)

$$\varepsilon_D = \frac{\%\Delta Q}{\%\Delta P} = \frac{\frac{\Delta Q}{Q_0}}{\frac{\Delta P}{P_0}} = \frac{\Delta Q}{\Delta P} \cdot \frac{P_0}{Q_0} \qquad \text{at a point: } \varepsilon_D = \text{slope} \times \frac{p}{Q}$$

$$\text{Rearranged: } \quad \%\Delta Q = \varepsilon_D \times \%\Delta P \qquad \%\Delta P = \frac{\%\Delta Q}{\varepsilon_D}$$

Example: $\varepsilon = -0.15$, $\%\Delta P = +70\% \Rightarrow \%\Delta Q = -0.15 \times 70 = -10.5\%$. Point: $\varepsilon = -2 \times \frac{360}{1280} = -0.56$

Elasticity ranges (use $|\varepsilon|$, no minus)

$\lvert\varepsilon\rvert = 0$	perfectly inelastic (vertical)
$0 < \lvert\varepsilon\rvert < 1$	inelastic
$\lvert\varepsilon\rvert = 1$	unit elastic
$\lvert\varepsilon\rvert > 1$	elastic
$\lvert\varepsilon\rvert = \infty$	perfectly elastic (horizontal)

Revenue

$$R = P \times Q \qquad \frac{R_1}{R_0} = \left(1 + \frac{\%\Delta P}{100}\right)\left(1 + \frac{\%\Delta Q}{100}\right)$$

	Price ↑	Price ↓
Inelastic	$R\uparrow$	$R\downarrow$
Elastic	$R\downarrow$	$R\uparrow$

Example: +10% price, −2% revenue: $\frac{0.98}{1.10} = 0.891 \Rightarrow \%\Delta Q = -10.9\%$

Linear demand $Q = a - bp$

$$\varepsilon_D = \frac{-bp}{a - bp} \qquad p_{\text{max revenue}} = \frac{a}{2b} \quad (\varepsilon = -1)$$

Example: $Q = 500 - 5p \Rightarrow p = \frac{500}{2 \times 5} = 50$

Other elasticities

$$\varepsilon_{XY} = \frac{\%\Delta Q_X}{\%\Delta P_Y} \qquad \varepsilon_I = \frac{\%\Delta Q}{\%\Delta \text{Income}} \qquad \varepsilon_S = \frac{\%\Delta Q_s}{\%\Delta P}$$

Cross: + substitutes, − complements.
Income: > 1 luxury, 0–1 necessity, < 0 inferior.

Two effects at once

$$\%\Delta Q_X = \varepsilon_{\text{own}} \times \%\Delta P_X \; + \; \varepsilon_{\text{cross}} \times \%\Delta P_Y$$

Example: Norway EVs: $-0.97 \times 20\% + 0.36 \times (-10\%) = -19.4\% - 3.6\% = -23\%$

Elasticity – rearranged: what do you know, what is asked?

You know	Asked for	Formula	Example
ε and $\%\Delta P$	$\%\Delta Q$	$\%\Delta Q = \varepsilon \times \%\Delta P$	$-0.15 \times 70\% = -10.5\%$ (London)
ε and the $\%\Delta Q$ you want	$\%\Delta P$ needed	$\%\Delta P = \dfrac{\%\Delta Q}{\varepsilon}$	$\dfrac{+25\%}{-1.2} = -20.8\%$ (club)
$\%\Delta Q$ and $\%\Delta P$	ε	$\varepsilon = \dfrac{\%\Delta Q}{\%\Delta P}$	$\dfrac{-10\%}{+25\%} = -0.4$ (restaurant)
$\%\Delta P$ and $\%\Delta$Revenue	ε	$\%\Delta Q = \dfrac{1+\%\Delta R}{1+\%\Delta P} - 1$, then $\varepsilon = \dfrac{\%\Delta Q}{\%\Delta P}$	$\dfrac{0.98}{1.10} - 1 = -10.9\%$; $\dfrac{-10.9}{10} = -1.09$
ε and $\%\Delta P$	$\%\Delta$Revenue	$\dfrac{R_1}{R_0} = (1+\%\Delta P) \times (1 + \varepsilon \times \%\Delta P)$	$1.10 \times 0.94 = 1.034 \Rightarrow +3.4\%$ (oil)
old value and $\%\Delta$	new value	$\text{new} = \text{old} \times \left(1 + \dfrac{\%\Delta}{100}\right)$	$300 \times 2.25 = 675$; $200 \times 0.895 = 179$
target quantity, ε, old price	new price	$\%\Delta P = \dfrac{\%\Delta Q}{\varepsilon}$, $P_1 = P_0 \times (1 + \%\Delta P)$	$40 \times (1 - 0.208) = 31.67$ (club)
two effects, one price change unknown	$\%\Delta P_Y$	$\%\Delta P_Y = \dfrac{\text{target } \%\Delta Q_X - \varepsilon_{\text{own}} \times \%\Delta P_X}{\varepsilon_{\text{cross}}}$	$\dfrac{12\% - 7\%}{0.15} = 33.3\%$ (Montana)
$\%\Delta Q_s$ and $\%\Delta P$	ε_S	$\varepsilon_S = \dfrac{\%\Delta Q_s}{\%\Delta P}$	$\dfrac{-20\%}{-20\%} = +1$ (2023, Q3.4)
ε_D, ε_S and $\%\Delta w$	new L_d, L_s	$\%\Delta L_d = \varepsilon_D \times \%\Delta w$, $\%\Delta L_s = \varepsilon_S \times \%\Delta w$	$-2 \times 25\% = -50\%$; $1.5 \times 25\% = +37.5\%$
equation $Q = a - bp$ and a point	ε at that point	$\varepsilon = -b \times \dfrac{p}{Q}$	$-2 \times \dfrac{360}{1280} = -0.56$
equation $Q = a - bp$	price where $\varepsilon = -1$	$p = \dfrac{a}{2b}$	$\dfrac{500}{2 \times 5} = 50$

Sign check: price ↓ and $\varepsilon < 0 \Rightarrow \%\Delta Q$ must be **positive**. Always use decimals inside brackets: $+25\% = 0.25$, $-10.5\% = -0.105$.

Consumer surplus (CS)

$$\text{CS} = \tfrac{1}{2} \times \left(p_{\max} - p^*\right) \times Q^*$$

$p_{\max}$ = price where demand hits the price axis (set $Q = 0$).
Table: CS = sum of (WTP − price) for units bought.

Example: $\tfrac{1}{2}(1000 - 360) \times 1280 = 409{,}600$

Producer surplus (PS)

Supply starts on the price axis (triangle):

$$\text{PS} = \tfrac{1}{2} \times \left(p^* - p_{\min}\right) \times Q^*$$

Supply starts on the quantity axis (trapezoid):

$$\text{PS} = \frac{Q_s(0) + Q^*}{2} \times p^*$$

Example: $\dfrac{200 + 1280}{2} \times 360 = 266{,}400$

Costs

$$\text{MC} = \Delta\,\text{TVC} \qquad \text{PS} = \text{Revenue} - \text{Variable costs}$$

$$\text{Profit} = \text{PS} - \text{Fixed costs}$$

Firm makes a unit if $P \geq \text{MC}$. Fixed costs don't affect supply.

Shifts

D↑ S↑ $Q \uparrow$, P ?
D↑ S↓ $P \uparrow$, Q ?
D↓ S↑ $P \downarrow$, Q ?
D↓ S↓ $Q \downarrow$, P ?

Market demand: add the quantities of all groups still buying at each price.

Topic 3 · Market Interventions I

Price ceiling (maximum price, below p^*)

Traded quantity $= Q_s$ (shortage $= Q_d - Q_s$). Find WTP = price on the demand curve at Q_s.

$$\text{CS} = (\text{WTP} - p_{\max}) \times Q_s + \tfrac{1}{2}(p_{\text{choke}} - \text{WTP}) \times Q_s \qquad \text{DWL} = \tfrac{1}{2}(\text{WTP} - p_{\max}) \times (Q^* - Q_s)$$

Example: $Q_d = 600 - 4p$, $Q_s = 100 + 6p$, ceiling 20: $Q_s = 220$, WTP $= \dfrac{600 - 220}{4} = 95$, CS $= \dfrac{(150 - 20) + (95 - 20)}{2} \times 220 =$ 22,550

Price floor / quota

Floor (above p^*): traded quantity $= Q_d$, surplus $= Q_s - Q_d$.
Quota: price read from the **demand** curve at the quota.

$$\text{PS} = (p - c) \times Q + \tfrac{1}{2}\, c \times Q$$

(c = supply price at that Q)

Example: Apples, quota 80: $p = 60$, $c = 26.7$, PS = 3,733

Minimum wage

Job destruction $= L_e - L_d$ Unemployment $= L_s - L_d$

With elasticities:

$$\%\Delta L_d = \varepsilon_D \times \%\Delta w \qquad \%\Delta L_s = \varepsilon_S \times \%\Delta w$$

Example: +25%: $L_d = 2000 \times 0.5 = 1000$, $L_s = 2000 \times 1.375 = 2750$, unemployment $= 1750$

Externalities

$$\text{Social cost} = \text{Private cost} + \text{External cost} \qquad \text{Social benefit} = \text{Private benefit} + \text{External benefit}$$

Efficient if social benefit $\geq$ social cost. Negative externality $\Rightarrow$ too much (tax). Positive $\Rightarrow$ too little (subsidy).

Example: Roger: $100 + 30 = 130 > 120 \Rightarrow$ should attend. Battery: private gain 1 < external cost 2 $\Rightarrow$ shouldn't happen.